Supplementary Information for: A Measured Communication-Quality Frontier for Federated LoRA Fine-Tuning with Adaptive Phase-Switching

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Table S1 Downstream zero-shot accuracy, TinyLlama-1.1B, single training seed (42), MPS corpus, 500 examples per benchmark. MMLU omitted: base accuracy is at chance for 4-way multiple choice

Method	ARC-Easy	BoolQ	HellaSwag	Δ vs. base (BoolQ)
Base (no adapter)	0.274	0.626	0.448	–
FLoRA	0.246	0.562	0.420	−6.4 pp
Two-Phase K=8	0.248	0.560	0.420	−6.6 pp
ReverseAdaptive	0.252	0.552	0.420	−7.4 pp
Spread (max–min)	0.006	0.010	0.000	–

Table S2 Downstream zero-shot accuracy, LLaMA-3.2-3B. Mean ± 1 standard deviation across three random seeds (42, 123, 456). 500 examples per benchmark. Cross-method spread is at most 0.5 percentage points on any benchmark and the per-method error bars overlap

Method	MMLU	ARC-Easy	BoolQ	HellaSwag
Base (no adapter)	0.544	0.832	0.706	0.540
FLoRA	0.558 ± 0.007	0.836 ± 0.007	0.736 ± 0.007	0.535 ± 0.003
Two-Phase K=8	0.558 ± 0.002	0.837 ± 0.008	0.738 ± 0.010	0.539 ± 0.004
ReverseAdaptive	0.557 ± 0.003	0.839 ± 0.008	0.739 ± 0.004	0.538 ± 0.004

Table S3 ReverseAdaptive threshold sensitivity (TinyLlama IID, seed 42, MPS corpus)

τ	Switch round	Final loss	Total comm. (MB)	Savings
0.001	11	1.2638	2083.13	19.20%
0.002	9	1.2675	1863.13	27.73%
0.005–0.2	6	1.2736	1533.13	40.53%

Algorithm S1 Bidirectional B-only protocol (server, per round)

Require: round r, phase boundary K, frozen-A cache (initially empty)
Ensure: upload mode, broadcast mode
if $r \leq K$ **then** ▷ FLoRA phase
  upload $\leftarrow$ full state (A, B)
  broadcast $\leftarrow$ full state
else if $r = K + 1$ **then** ▷ transition: seed frozen A
  upload $\leftarrow$ full state (A, B)
  $A_{\text{frozen}} \leftarrow A$ from the first client's uploaded state
  broadcast $\leftarrow B$ only ▷ clients keep round-K A
else ▷ FFA-LoRA steady state
  upload $\leftarrow B$ only ▷ server prepends A_{frozen}
  broadcast $\leftarrow B$ only ▷ clients keep A_{frozen}
end if

Table S4 Experimental setup

Parameter	Value
Base models	TinyLlama-1.1B-Chat-v1.0 [1]; LLaMA-3.2-3B [2]
Datasets	Alpaca [3]; Dolly-15k [4]; 3000 samples each
Data partitions	IID; Dirichlet α=0.5; Dirichlet α=0.1
Clients / round	10 / 10 (full participation)
Rounds	15
Local epochs	1
Batch size	4 (TinyLlama), 2 (LLaMA-3.2-3B)
Grad. accumulation	4 (TinyLlama), 8 (LLaMA-3.2-3B); effective batch 16
Learning rate	10^{-4} (AdamW [5])
Max sequence length	256
LoRA rank / alpha	16 / 32
LoRA target modules	`q_proj`, `v_proj`
LoRA dropout	0.1
Hardware (primary)	Apple M4 Pro, 48 GB unified memory, MPS
Hardware (reproduction)	NVIDIA RTX 4090 (TinyLlama); A100 40 GB (LLaMA-3.2-3B)
Training dtype	float32 (TinyLlama-1.1B); float16 base + float32 LoRA (LLaMA-3.2-3B)
Transmitted dtype	float32 (TinyLlama-1.1B); float16 (LLaMA-3.2-3B)
Downstream eval	500 examples/benchmark, seed 42, log-likelihood
Held-out eval	500 examples, train indices 3001–3500, seq. len. 256
Benchmarks	MMLU [6], ARC-Easy [7], BoolQ [8], HellaSwag [9]
Seeds (TinyLlama IID)	{42, 123, 456}
Seeds (LLaMA-3.2-3B)	{42, 123, 456}
ReverseAdaptive defaults	`warmup_rounds`=5, τ=0.01, `stability_threshold`=1.1, `transition_rounds`=0 (instantaneous)

Supplementary Tables and Algorithm

Supplementary Note 1

All experiments use AdamW [5] with learning rate 10^{-4}, β_1=0.9, β_2=0.999, and weight decay 0.01 (the PyTorch default). No learning rate schedule is applied. Gradients are

clipped to a maximum ℓ_2 norm of 1.0 at every optimizer step. Maximum sequence length is 256 tokens. See Table S4 for the complete configuration including ReverseAdaptive defaults.

S0.1 Reproducibility Checklist

Code: Full source code is publicly released, including all configuration YAML files, training scripts, evaluation scripts, and scripts to reproduce every figure and table from the analysis CSVs.

Verification: A script in the repository (`scripts/verify_numbers.py`) recomputes the paper's principal quantitative claims from the released analysis files and run records with 179 automated assertions.

Seeds: The IID seeds are listed in Table S4. The α=0.1 partition-sensitivity runs additionally use seeds 789 and 1000, which appear only in the partition-sensitivity results of Supplementary Note 2, alongside the other per-seed results.

Hardware: The MPS corpus was executed on a single Apple M4 Pro workstation with 48 GB unified memory; a core of 34 of its runs, approximately 285 hours of compute, was re-executed on commercially rented NVIDIA hardware, an RTX 4090 for TinyLlama-1.1B and an A100 40 GB for LLaMA-3.2-3B; the comparison is in Supplementary Note 4. The Dolly-15k replication and the FFA-LoRA and FedIT baselines were produced on the rented hardware only.

Data: Alpaca [3] via `tatsu-lab/alpaca` on HuggingFace, 3000-sample subset, and Dolly-15k [4] via `databricks/databricks-dolly-15k`, matched subset. Adapter checkpoints are not released, except the final adapter states of the three seed-42 TinyLlama-1.1B runs evaluated in Table S1.

S0.2 Disabling the Switch

Running ReverseAdaptive with switching disabled reproduces the matched FLoRA [10] run to 10 decimal places at every round, establishing that the wrapper is a zero-cost modification when unused: it rules out numerical artifacts from the wrapper's presence, not merely from its activation. The comparison is within a single backend and a single execution environment. Agreement across hardware is weaker than bit-identity and is reported in Supplementary Note 4.

Disabling the switch requires care. Setting τ to a small positive value does not achieve it: any round in which the training loss fails to improve satisfies $\rho_r < \tau$, so the criterion fires. On the FLoRA trajectory underlying the threshold ablation the loss rises once, at round 11, so τ=0 still triggers a switch at that round, and suppressing the switch entirely requires τ below the most negative round-over-round relative change observed, approximately -0.0009 here. The sanity baseline accordingly uses τ=-1.0 together with a warmup exceeding the round budget. A practitioner wanting the wrapper present but inert should disable it by one of those means rather than by choosing a conservative threshold.

Table S5 Per-seed TinyLlama-1.1B IID results, CUDA corpus, backing main text Table 1. Communication is deterministic across seeds. For Two-Phase the Switch/boundary column shows the predetermined phase boundary K, not an adaptive switch event; for FFA-LoRA, A is frozen from initialization (shown as "init"); for ReverseAdaptive it shows the round at which the plateau signal triggered

Method	Seed	Final loss	Total MB	Switch / boundary
FLoRA	42	1.261408	2578.13	–
FLoRA	123	1.260479	2578.13	–
FLoRA	456	1.260500	2578.13	–
FedIT	42	1.260767	2578.13	–
FedIT	123	1.258420	2578.13	–
FedIT	456	1.261503	2578.13	–
Two-Phase K=8	42	1.271276	1863.13	K=8
Two-Phase K=8	123	1.270130	1863.13	K=8
Two-Phase K=8	456	1.270237	1863.13	K=8
ReverseAdaptive (τ=0.01)	42	1.275514	1533.13	6
ReverseAdaptive (τ=0.01)	123	1.274610	1533.13	6
ReverseAdaptive (τ=0.01)	456	1.274481	1533.13	6
FFA-LoRA	42	1.303975	983.13	init
FFA-LoRA	123	1.301738	983.13	init
FFA-LoRA	456	1.303628	983.13	init

Table S6 Non-IID results, MPS corpus, five seeds at α=0.1. This table documents partition sensitivity rather than comparing methods, so the larger MPS seed set is retained; Table S5 follows the CUDA corpus because it backs a method comparison. Seeds 42 and 1000 at α=0.1 had 9 and 8 active clients due to empty Dirichlet partitions, which scales communication to 9/10 and 8/10 of the full value (1379.81 and 1226.50 MB against 1533.13). Cross-backend verification (Supplementary Note 4) covers seeds 42, 123, and 456; seeds 789 and 1000 were executed on MPS only

Setting	Seed	Final loss	Total MB	Switch round	Active clients
ReverseAdaptive, α=0.5	42	1.285933	1533.13	6	10
ReverseAdaptive, α=0.5	123	1.282625	1533.13	6	10
ReverseAdaptive, α=0.5	456	1.252133	1533.13	6	10
ReverseAdaptive, α=0.1	42	1.268353	1379.81	6	9
ReverseAdaptive, α=0.1	123	1.288985	1533.13	6	10
ReverseAdaptive, α=0.1	456	1.348083	1533.13	6	10
ReverseAdaptive, α=0.1	789	1.259200	1533.13	6	10
ReverseAdaptive, α=0.1	1000	1.273046	1226.50	6	8

Supplementary Note 2

Tables S5 to S8 give per-seed results and are labeled with their corpus, which follows the corresponding table in the main text. The TinyLlama-1.1B method comparisons use the CUDA corpus, since the FFA-LoRA, FedIT, and Dolly-15k runs exist only there; the LLaMA-3.2-3B results use MPS, matching main text Table 3; and the partition-sensitivity results retain the larger MPS seed set, where five seeds at α=0.1 provide better evidence than three.

At α=0.5, mean final loss is statistically indistinguishable from IID for every method in the released analysis CSVs, with differences of 0.001 to 0.007 against per-seed standard deviations near 0.02 over three seeds. Heterogeneity manifests instead as

Table S7 Per-seed LLaMA-3.2-3B IID results, MPS corpus, backing main text Table 3. ReverseAdaptive uses τ=0.01. Communication is deterministic given the switch round; the three ReverseAdaptive seeds occupy three adjacent lattice points

Method	Seed	Final loss	Total MB	Switch
FLoRA	42	1.371683	2625.0	–
FLoRA	123	1.373402	2625.0	–
FLoRA	456	1.371998	2625.0	–
Two-Phase K=8	42	1.393776	1942.5	K=8
Two-Phase K=8	123	1.394877	1942.5	K=8
Two-Phase K=8	456	1.392772	1942.5	K=8
ReverseAdaptive	42	1.393831	1837.5	8
ReverseAdaptive	123	1.390265	1942.5	9
ReverseAdaptive	456	1.397366	1732.5	7

Table S8 Per-seed TinyLlama-1.1B Dolly-15k IID results, CUDA corpus, backing main text Table 2. Held-out Δ is tuned minus base on the Dolly-15k held-out slice; more negative is better

Method	Seed	Final loss	Held-out Δ	Total MB
FLoRA	42	1.656762	−0.531848	2578.13
FLoRA	123	1.652934	−0.533376	2578.13
FLoRA	456	1.653426	−0.533664	2578.13
Two-Phase K=8	42	1.666616	−0.527571	1863.13
Two-Phase K=8	123	1.662685	−0.529113	1863.13
Two-Phase K=8	456	1.663475	−0.529410	1863.13
ReverseAdaptive	42	1.671039	−0.526568	1533.13
ReverseAdaptive	123	1.666995	−0.527418	1533.13
ReverseAdaptive	456	1.667699	−0.526547	1533.13

sensitivity to the partition draw: seed-to-seed standard deviation rises by roughly an order of magnitude, from 0.5 to 1.6×10^{-3} under IID in Table S5 to 18 to 25×10^{-3} at α=0.5 across the methods in the released CSVs, which is also why the per-seed spread at α=0.1 in Table S6 is wide.

Supplementary Note 3

Table S9 reports paired t-tests over three seeds on final-round training loss and on held-out instruction-following loss. TinyLlama-1.1B results use the CUDA corpus, matching main text Table 1; LLaMA-3.2-3B results use the MPS corpus.

With three seeds these tests have very low power, so effect sizes should be read alongside the p-values. No multiplicity correction is applied in the main results tables. Applying a Bonferroni correction across the six TinyLlama-1.1B Alpaca IID comparisons gives a threshold of 0.0083, which the ReverseAdaptive vs. Two-Phase K=8 comparison on held-out loss ($p = 0.026$) does not meet. That comparison is secondary under the frontier framing of main text Discussion; the load-bearing comparison is

Table S9 Paired t-tests, three seeds. TinyLlama-1.1B rows use the CUDA corpus; LLaMA-3.2-3B rows use MPS. Positive Δ indicates the first method has higher loss. Held-out instruction-following loss was not computed for the LLaMA-3.2-3B corpus; quality at that scale is reported as final training loss and downstream zero-shot accuracy

Comparison	Δ final loss	p	Δ held-out	p
TinyLlama-1.1B, Alpaca, IID				
Two-Phase K=8 vs. FLoRA	+0.0098	4.2×10^{-5}	+0.0034	7.9×10^{-5}
ReverseAdaptive vs. FLoRA	+0.0141	1.1×10^{-5}	+0.0063	6.2×10^{-3}
ReverseAdaptive vs. Two-Phase K=8	+0.0043	3.4×10^{-4}	+0.0029	2.6×10^{-2}
FFA-LoRA vs. ReverseAdaptive	+0.0282	4.4×10^{-4}	+0.0182	3.8×10^{-4}
FFA-LoRA vs. FLoRA	+0.0423	1.7×10^{-4}	+0.0245	1.1×10^{-4}
FedIT vs. FLoRA	−0.0006	0.588	+0.0002	0.425
TinyLlama-1.1B, Dolly-15k, IID				
Two-Phase K=8 vs. FLoRA	+0.0099	7.8×10^{-5}	+0.0043	2.4×10^{-6}
ReverseAdaptive vs. FLoRA	+0.0142	2.5×10^{-5}	+0.0061	7.6×10^{-3}
LLaMA-3.2-3B, Alpaca, IID				
Two-Phase K=8 vs. FLoRA	+0.0214	3.2×10^{-4}	–	–
ReverseAdaptive vs. FLoRA	+0.0215	1.3×10^{-2}	–	–
ReverseAdaptive vs. Two-Phase K=8	+0.000012	0.997	–	–

Table S10 Cross-backend verification summary, MPS against CUDA, 34 run pairs

Category	Runs
Agreed within setting-aware tolerance	27
Disagreed	4
Comparable pairs	31
No valid MPS reference (CUDA reported)	3
Total	34

ReverseAdaptive vs. FFA-LoRA, which holds at $p < 10^{-3}$ on both metrics with a difference roughly twenty times the largest per-method standard deviation.

The LLaMA-3.2-3B comparison between ReverseAdaptive and Two-Phase K=8 has a 95% confidence interval of $[-0.0114, 0.0114]$, which contains the 0.0043 difference observed between the same methods at TinyLlama-1.1B. The test therefore cannot distinguish equivalence from a difference of the magnitude seen at the smaller scale.

Supplementary Note 4

The primary corpus was produced on the MPS backend and subsequently re-executed on CUDA. Table S10 summarizes the comparison; Table S11 details every disagreement.

Tolerances are setting-aware. IID runs require per-round training loss to agree within 0.01 absolute. Non-IID runs allow 0.025, since heterogeneous partitions amplify per-round variation, but additionally require the switch round to match exactly and permit no more than two consecutive rounds exceeding 0.01. Communication must agree within 0.01 in all settings, which is effectively exact since totals are protocol-deterministic.

Of 34 run pairs, 31 have a valid MPS reference. The three Two-Phase K=8 runs at α=0.5 do not: their MPS executions predate the bidirectional B-only implementation

Table S11 Cross-backend disagreements. Every communication difference is one 110 MB lattice step. Run 16 failed only the consecutive-spike rule

Run	Config.	MPS	CUDA	Note
11	τ=0.001, IID	2083.13	1973.13	s: 11 to 10
12	τ=0.002, IID	1863.13	1753.13	s: 9 to 8
16	α=0.1, 123	1533.13	1533.13	spike rule
17	α=0.1, 456	1533.13	1643.13	s: 6 to 7

and never switched, recording 2578.13 MB, the full FLoRA volume. Those are absent references rather than disagreements, and CUDA is the source of truth for that configuration. Of the 31 comparable pairs, 27 agree within tolerance and 4 do not.

Three of the four disagreements are the same phenomenon: every observed cross-backend communication difference is exactly one 110 MB switch-round step (main text Discussion), so Table S11 lists adjacent lattice points. The backends agree on the protocol and disagree only on which round the plateau fires, and only near a boundary: two tight-τ IID runs (τ=0.001, τ=0.002) and one α=0.1 run. Run 16 has no communication or switch-round difference: only the consecutive-spike rule failed.

Run 17 was re-executed on CUDA with partition seed 456 and training seed 999 and again switched at round 7, so the one-round gap is a stable backend/partition property rather than run noise. At α=0.1, exact switch-round agreement is therefore not guaranteed (one of three CUDA counterparts fires one round later); CUDA is reported as primary. The switch round matched exactly for every IID run at τ=0.01 and for all three LLaMA-3.2-3B ReverseAdaptive runs, supporting the cross-scale claim in main text Discussion.

Byte accounting. Transport-layer counts use `numel`×`element_size` on tensors actually sent; the FFA-LoRA `get_communication_cost` helper is unused.

Code revisions. Runs span multiple revisions (including uncommitted working-tree changes), so numbers come from released records and the replication here. Byte totals are revision-invariant: FLoRA, Two-Phase K=8 and ReverseAdaptive each matched under two CUDA revisions at 2578.13, 1863.13 and 1533.13 MB. Loss carries revision uncertainty; the FLoRA and FedIT rows of main text Table 1 differ by 0.0006 across that boundary, against a ReverseAdaptive-to-FFA-LoRA gap of 0.0282.

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